Crank–Nicolson, the θ-scheme, and Rannacher time-stepping

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1 Setting and motivation

FTCS and BTCS (previous two notes) are first-order in time: their local truncation error is $O(\Delta\tau)+O(\Delta x^2)$. To balance the spatial and temporal contributions one needs $\Delta\tau \sim \Delta x^2$, which forces $M \sim N^2$ and total cost $O(N^3)$ for both schemes (FTCS by CFL, BTCS by accuracy).

This note implements *Crank–Nicolson* (CN), the unique scheme in the convex family of FTCS and BTCS that is *second-order in time*: $O(\Delta\tau^2) + O(\Delta x^2)$. With $M \sim N$, CN achieves $O(\Delta x^2)$ precision at $O(N^2)$ total cost — a factor-N improvement over FTCS/BTCS in the same precision regime. CN is the workhorse of finite-difference option pricers in production.

The deliverable is a θ-scheme infrastructure (`quantlib.theta_scheme`) that subsumes FTCS, BTCS and CN as parameter choices, plus the CN pricer with Rannacher time-stepping (`quantlib.cn`) to suppress spurious oscillations near the strike kink.

2 The θ-family

The three schemes FTCS, BTCS and CN are particular cases of a single parametric family:

$$\frac{V_j^{n+1} - V_j^n}{\Delta\tau} \;=\; \theta\,\mathcal{L}_h V_j^{n+1} \;+\; (1-\theta)\,\mathcal{L}_h V_j^n, \qquad \theta \in [0,1], \tag{1}$$

where $\mathcal{L}_h$ is the discrete spatial operator

$$\mathcal{L}_h V_j \;=\; \frac{\sigma^2}{2}\,\frac{V_{j+1} - 2V_j + V_{j-1}}{\Delta x^2} + \mu\,\frac{V_{j+1} - V_{j-1}}{2\Delta x} - r\,V_j.$$

The cases of interest are

θ	Name	Type	Stability	Time order
0	FTCS	Explicit	Conditional ($\alpha \le 1/2$)	$O(\Delta\tau)$
1/2	**Crank–Nicolson**	Implicit	Unconditional	$O(\Delta\tau^2)$
1	BTCS	Implicit	Unconditional	$O(\Delta\tau)$

The order jump at $\theta = 1/2$ is the central observation of this note. It is the pay-off of the symmetry between the new and old levels: when the right-hand side is the average of $\mathcal{L}_h V$ at τ_n and τ_{n+1}, the discrete equation is centred at the *midpoint* $\tau_{n+1/2}$, and the midpoint rule is one order more accurate than the endpoint rules.

3 Crank–Nicolson explicitly

Setting $\theta = 1/2$ in (1), expanding $\mathcal{L}_h$, and rearranging V^{n+1} to the left, V^n to the right:

$$\boxed{\beta_-\,V_{j-1}^{n+1} + \beta_0\,V_j^{n+1} + \beta_+\,V_{j+1}^{n+1} \;=\; \gamma_-\,V_{j-1}^n + \gamma_0\,V_j^n + \gamma_+\,V_{j+1}^n.} \tag{2}$$

With $\alpha = \frac{\sigma^2}{2}\,\Delta\tau/\Delta x^2$ and $\nu = \mu\,\Delta\tau/(2\Delta x)$ as before, the LHS coefficients are

$$\beta_- = -\tfrac{1}{2}\alpha + \tfrac{1}{2}\nu, \quad \beta_0 = 1 + \alpha + \tfrac{1}{2}r\Delta\tau, \quad \beta_+ = -\tfrac{1}{2}\alpha - \tfrac{1}{2}\nu,$$

and the RHS coefficients are

$$\gamma_- = +\tfrac{1}{2}\alpha - \tfrac{1}{2}\nu, \quad \gamma_0 = 1 - \alpha - \tfrac{1}{2}r\Delta\tau, \quad \gamma_+ = +\tfrac{1}{2}\alpha + \tfrac{1}{2}\nu.$$

Symmetry: in terms of the BTCS coefficients $b_\pm, b_0$ and the FTCS coefficients $a_\pm, a_0$ of the previous notes, $\beta_\pm = \frac{1}{2}b_\pm$, $\beta_0 - 1 = \frac{1}{2}(b_0 - 1)$, and $\gamma_\pm = \frac{1}{2}a_\pm$, $\gamma_0 - 1 = \frac{1}{2}(a_0 - 1)$. CN is, structurally, "half BTCS plus half FTCS".

4 Why CN is second-order in time: the midpoint rule

The non-trivial claim is that the local truncation error of (2) is $O(\Delta\tau^2) + O(\Delta x^2)$, despite the fact that the time difference $(V^{n+1} - V^n)/\Delta\tau$ looks like the same first-order forward Euler quotient that FTCS uses. The gain comes from *which time level the equation is centred on.*

Expand V around the midpoint $\tau_{n+1/2} = \tau_n + \Delta\tau/2$:

$$V(\tau_{n+1}) = V_* + \tfrac{1}{2}\Delta\tau\, V_*' + \tfrac{1}{8}\Delta\tau^2\, V_*'' + \tfrac{1}{48}\Delta\tau^3\, V_*''' + O(\Delta\tau^4),$$
$$V(\tau_n) = V_* - \tfrac{1}{2}\Delta\tau\, V_*' + \tfrac{1}{8}\Delta\tau^2\, V_*'' - \tfrac{1}{48}\Delta\tau^3\, V_*''' + O(\Delta\tau^4),$$

where $V_* := V(\tau_{n+1/2})$, $V_*' := V_\tau(\tau_{n+1/2})$, etc. Subtracting and dividing by $\Delta\tau$:

$$\frac{V(\tau_{n+1}) - V(\tau_n)}{\Delta\tau} \;=\; V_\tau(\tau_{n+1/2}) \;+\; \frac{\Delta\tau^2}{24}\, V_{\tau\tau\tau}(\tau_{n+1/2}) \;+\; O(\Delta\tau^4). \tag{3}$$

The $\Delta\tau$ term *cancels by symmetry*: the quotient is a second-order approximation to V_τ at the midpoint.

Analogously, adding the two expansions of $\mathcal{L}_h V$ (one at τ_n, one at τ_{n+1}) and dividing by 2:

$$\tfrac{1}{2}[\mathcal{L}_h V(\tau_{n+1}) + \mathcal{L}_h V(\tau_n)] \;=\; \mathcal{L}_h V(\tau_{n+1/2}) \;+\; O(\Delta\tau^2) \;+\; O(\Delta x^2), \tag{4}$$

where the $O(\Delta\tau^2)$ comes from the time interpolation (linear average of the endpoints) and the $O(\Delta x^2)$ is the spatial truncation of $\mathcal{L}_h$. Combining (3) and (4), the CN equation $(V^{n+1} - V^n)/\Delta\tau = \frac{1}{2}(\mathcal{L}_h V^{n+1} + \mathcal{L}_h V^n)$ is the discretisation of the exact PDE $V_\tau = \mathcal{L}V$ *at the midpoint*, with truncation

$$\tau_h = O(\Delta\tau^2) + O(\Delta x^2).$$

FTCS and BTCS, by contrast, evaluate the right-hand side at τ_n or τ_{n+1} (not the midpoint); the time error there is $O(\Delta\tau)$.

5 The linear system and matrix structure

Equation (2) written for the interior nodes $j = 1, \ldots, N-1$ becomes

$$A\,\mathbf{V}_{\text{int}}^{n+1} \;=\; B\,\mathbf{V}_{\text{int}}^{n} + \mathbf{b}_{\text{bnd}}^{n},$$

where A and B are tridiagonal $(N-1)\times(N-1)$ matrices with $(\beta_-, \beta_0, \beta_+)$ and $(\gamma_-, \gamma_0, \gamma_+)$ on their diagonals, and the boundary vector mixes both time levels:

$$\mathbf{b}_{\text{bnd}}^{n} = \begin{pmatrix} \gamma_-\, V_0^n - \beta_-\, V_0^{n+1} \\ 0 \\ \vdots \\ 0 \\ \gamma_+\, V_N^n - \beta_+\, V_N^{n+1} \end{pmatrix}. \tag{5}$$

Both lateral coefficients of the stencils contribute to the boundary correction, in contrast to BTCS where only the LHS coefficients appeared.

The per-step algorithm has three phases:

1. Apply the RHS stencil B to $\mathbf{V}_{\text{int}}^n$ as a three-point stencil sweep ($O(N)$, no need to assemble B explicitly).
2. Add the boundary correction $\mathbf{b}_{\text{bnd}}^n$ ($O(1)$: only two endpoints are non-zero).
3. Solve $A\,\mathbf{V}_{\text{int}}^{n+1} = (\text{result})$ by Thomas with the pre-factored A ($O(N)$).

Total cost per step: $O(N)$. Total cost: $O(N \cdot M)$, the same as BTCS.

Pre-factoring A. The matrix A is constant in time (again thanks to the constant-coefficient form of the transformed PDE). Factoring A once at setup and applying the factor at every step is the standard pattern, identical to BTCS.

6 Stability of the θ-scheme

The von Neumann amplification factor of (1), ignoring the reactive term, is

$$g(\xi) = \frac{1 - 4\alpha(1-\theta)\sin^2(\xi/2) + i\,(1-\theta)\beta_C \sin\xi}{1 + 4\alpha\theta\sin^2(\xi/2) - i\,\theta\beta_C \sin\xi}, \tag{6}$$

with $\beta_C := \mu\Delta\tau/\Delta x$, $\xi \in [-\pi, \pi]$ the Fourier wavenumber.

Proposition 6.1 (Stability of the θ-scheme)**.** *The θ-scheme is unconditionally von Neumann stable if and only if $\theta \geq 1/2$. For $\theta < 1/2$, $|g| \leq 1$ requires the conditional bound*

$$\alpha \;\leq\; \frac{1}{2(1-2\theta)}.$$

In particular, $\theta = 0$ gives $\alpha \leq 1/2$ (the FTCS CFL condition), and $\theta = 1/2$ is the smallest θ for which the bound is vacuous.

Sketch. Set $s = \sin^2(\xi/2)$, $u = \sin\xi$, so $u^2 = 4s(1-s)$. Then $|g|^2 = N(s,u)/D(s,u)$ where both numerator and denominator are quadratic in α. The condition $|g|^2 \leq 1$ becomes $D - N \geq 0$, which after expansion is

$$8\alpha s\left[1 + 2\alpha(2\theta - 1)s\right] \;+\; (2\theta - 1)\,\beta_C^2 u^2 \;\geq\; 0.$$

For $\theta \geq 1/2$, the bracket is at least 1 and the β_C term is non-negative, for any $\alpha \geq 0$, $s \in [0,1]$. For $\theta < 1/2$, take the Nyquist mode $s = 1$, $u = 0$: the condition becomes $1 - 2\alpha(1-2\theta) \geq 0$, which is the stated inequality. (Full algebra in Strikwerda, Chapter 6, or [4].) □ □

CN at $\theta = 1/2$ is therefore *exactly on the boundary* between conditional and unconditional stability. This is the structural reason why CN, although stable, has weaker damping properties than BTCS, as we now examine.

7 The kink-induced oscillation problem

The amplification factor at the Nyquist wavenumber $\xi = \pi$ (highest representable frequency on the grid, $\sin^2(\pi/2) = 1$, $\sin\pi = 0$) is

$$g_\theta(\pi) = \frac{1 - 4\alpha(1-\theta)}{1 + 4\alpha\theta}.$$

For BTCS ($\theta = 1$) and CN ($\theta = 1/2$):

$$g_{\mathrm{BTCS}}(\pi) = \frac{1}{1+4\alpha} \xrightarrow{\alpha\to\infty} 0, \qquad g_{\mathrm{CN}}(\pi) = \frac{1-2\alpha}{1+2\alpha} \xrightarrow{\alpha\to\infty} -1.$$

BTCS damps the Nyquist mode exponentially; CN preserves it with a sign flip. In the regime of interest for CN ($\Delta\tau \sim \Delta x$, so $\alpha \sim 1/\Delta x \to \infty$ as $N \to \infty$), $|g_{\mathrm{CN}}(\pi)|$ is essentially 1, and the Nyquist mode is preserved indefinitely while alternating sign in time.

This is harmless for smooth initial data: the Fourier coefficient of the Nyquist mode in the initial vector V^0 is exponentially small, and "preserving exponentially small" yields nothing visible.

The payoff is not smooth. A vanilla call payoff $\phi(x) = K(e^x - 1)^+$ has a kink at $x = 0$: ϕ is continuous but ϕ' has a jump. The Fourier expansion of a function with such a kink has coefficients decaying only as $1/k^2$, not exponentially. The Nyquist coefficient in V^0 is therefore $O(1/N^2)$ — not exponentially small.

Consequence. An $O(1/N^2)$ amplitude at the Nyquist mode, preserved across M time steps with sign flip $g \approx -1$ at each, produces a visible oscillation pattern in the computed V^M:

- Bounded ($|g| \le 1$, so the oscillation does not grow).
- Persistent (each step preserves the magnitude).
- Localised near the kink (where the kink-induced mode lives).
- Alternating sign at consecutive grid nodes (the spatial signature of $\xi = \pi$).

The visible artefact is a sawtooth of small amplitude near the strike, distinct from the explosive sawtooth of FTCS instability (see the FTCS note) by being bounded but visible. How much it degrades the price depends on $\Delta\tau/\Delta x$. With $\Delta\tau$ comparable to Δx the price at the strike is barely affected on the grids used here: for the at-the-money call of `validate_cn.py` ($S = K = 100$, $\sigma = 0.2$, $T = 1$) with $M = N/2$ ($\Delta\tau/\Delta x = 1.25$), vanilla CN converges with error ratio 4.00 per refinement (5.68×10^{-3} at $N = 100$ to 8.86×10^{-5} at $N = 800$), slightly more accurate than with Rannacher. The loss of order appears when $\Delta\tau \gg \Delta x$: test [3] of `validate_cn.py` ($N = 800$, $M = 10, 20, 40$) gives error ratios 2.11 and 2.53 under halving of $\Delta\tau$, against 3.89 and 3.55 with Rannacher. Giles and Carter [6] observe first-order convergence of the price at the largest $\Delta\tau/\Delta x$ they test, and note that for smaller ratios the same behaviour shows only on much finer grids. The Greeks (gamma in particular) inherit the oscillation.

8 Rannacher time-stepping

The standard fix goes back to Rannacher [5], who replaced the first CN time step by two half-time-step implicit (BTCS) steps. The version used here, from Giles and Carter [6], replaces the first *two* CN time steps by *four* half-time-step BTCS steps, each of length $\Delta\tau/2$:

$$\underbrace{\mathrm{BTCS}_{\Delta\tau/2} \circ \mathrm{BTCS}_{\Delta\tau/2} \circ \mathrm{BTCS}_{\Delta\tau/2} \circ \mathrm{BTCS}_{\Delta\tau/2}}_{\text{4 half-steps, total time } 2\Delta\tau} \circ \underbrace{\mathrm{CN}_{\Delta\tau} \circ \cdots \circ \mathrm{CN}_{\Delta\tau}}_{M-2 \text{ full steps}}. \tag{7}$$

Why it works. Each BTCS half-step damps the Nyquist mode by a factor $g_{\mathrm{BTCS},\Delta\tau/2}(\pi) = 1/(1+2\alpha)$ (α is the CN Fourier number; the half-step uses $\alpha/2$, giving $1/(1+2\alpha)$). After four half-steps the Nyquist mode is multiplied by $1/(1+2\alpha)^4$, which for $\alpha = O(N) \gg 1$ is $O(1/N^4)$ — well below the $O(\Delta x^2) = O(1/N^2)$ floor of the discretisation. By the time CN takes over, the kink-induced high-frequency modes are damped to round-off, and CN proceeds with effectively smooth data.

Why two full steps. Rannacher's original start-up (one CN step replaced by two implicit half-steps) already recovers second-order convergence in L^2. Giles & Carter [6] show that replacing two CN steps by four half-steps is needed for second-order accuracy in L^∞ for the value, delta and gamma; that is the protocol (7) and the default here.

Why BTCS, not finer CN. Refining $\Delta\tau$ does remove the artefact, but only once $\alpha = \sigma^2\Delta\tau/(2\Delta x^2)$ is $O(1)$. As $\Delta\tau \to 0$ at fixed Δx, $\alpha \to 0$ and $g_{\mathrm{CN}}(\pi) = (1-2\alpha)/(1+2\alpha) \to +1$ with no sign flip: over $[0, T]$ the Nyquist mode decays like $e^{-2\sigma^2 T/\Delta x^2}$, as it should. For $\alpha \le 1/2$ there is no sign flip at all, and for α of order one the flipping mode loses a factor $(2\alpha-1)/(2\alpha+1)$ per step ($1/3$ at $\alpha = 1$). The persistent sign-flipping mode needs $\alpha \gg 1$, i.e. $\Delta\tau$ large relative to Δx^2, which is exactly the regime $\Delta\tau \sim \Delta x$ in which CN is worth using. Taking $\Delta\tau \sim \Delta x^2$ instead would bring back $M \sim N^2$ and the $O(N^3)$ cost of FTCS/BTCS. Within the CN regime only a dissipative scheme can remove the irregular high-frequency content, and BTCS is the natural choice because the same Thomas factorisation can be reused with the appropriate coefficients.

Cost. The four BTCS half-steps replace two CN steps, so the protocol adds two tridiagonal solves, about $2/M$ of the CN cost, plus one extra factorisation for the half-step matrix: a noticeable overhead at $M = 10$, negligible by $M = 100$, and small compared to the convergence gain when $\Delta\tau \gg \Delta x$.

9 Implementation outline

The implementation consists of two modules:

- `quantlib.theta_scheme` (Python) / `quant::pde::theta_scheme_*` (C++): a generic θ-stepper parametrised by θ. Internally pre-factors the LHS matrix and runs a configurable number of time steps. Reusable as scaffolding for the American put (where the LHS solve is replaced by PSOR).
- `quantlib.cn` (Python) / `quant::pde::cn_*` (C++): the user-facing CN pricer for European options, with Rannacher smoothing enabled by default (`rannacher_steps=2`). Setting `rannacher_steps=0` gives vanilla CN, exposed to make the oscillation phenomenon testable.

  The signature mirrors BTCS:

```
cn_european_call(S, K, r, sigma, T, *, N, M, n_sigma=4.0,
                 rannacher_steps=2)
```

The θ-scheme module exposes a low-level kernel `theta_march(grid, V0, theta, bc_lower, bc_upper)` that the CN pricer calls twice for the Rannacher protocol: once with $\theta = 1$ for the initial half-stepped BTCS warm-up, then with $\theta = 1/2$ for the rest.

No replacement of FTCS/BTCS. The FTCS and BTCS modules do not use it. The θ-scheme infrastructure is a parallel mechanism intended for CN and any future generalisations.

10 Validation strategy

Six tests in `validate_cn.py` certify the new pricer (`test_cn.cpp` repeats the others for the C++ pricer; the fourth is in `test_theta_scheme.cpp`):

1. Cross-validation against Black–Scholes closed form on a parameter grid (the standard battery used for FTCS and BTCS): calls and puts on eight parameter sets at $N = 400$, $M = 200$, tolerance 5×10^{-3}.

2. **Quadratic convergence in time:** at $N = 800$, halving $\Delta\tau$ ($M = 10, 20, 40$) must reduce the error by a factor above 3, against ≈ 2 for BTCS; the measured ratios are 3.89 and 3.55. This is *the* test of CN.

3. **Vanilla CN (without Rannacher) has degraded convergence:** the same problem with `rannacher_steps=0` must give ratios below 3 (measured 2.11 and 2.53) and an error at $M = 10$ at least five times the Rannacher one. This is the empirical evidence of the kink-induced oscillation.

4. Reproduction of BTCS via $\theta = 1$ in the θ-scheme: `theta_march` with $\theta = 1$ matches the BTCS pricer to round-off.

5. Put–call parity at five spots.

6. Input validation ($S \leq 0$, negative `rannacher_steps`, `rannacher_steps` $> M$, spot outside the truncated domain).

The third test is the operational signature of the kink-induced oscillation, analogous to the CFL-violation diagnostic of FTCS: a configuration that produces visibly degraded output, captured as a passing assertion to prevent regression.

References

References

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